WORK-RELATED INJURIES

IN THE ITALIAN TRANSPORTATION AND STORAGE SECTOR

Exploratory Data Analysis

Author: Giulio Galbiati

Amazon Data Analytics Cohort 18

Table of Contents

Document Control 4
Document Status 4
Exploratory Data Analysis (EDA) 5
1. Executive Overview 6
1.1 Purpose 6
1.2 Analytical Roadmap 6
1.3 Key Questions 6
1.4 Headline Findings 6
1.5 Analytical Positioning 7
2. Data & Variables Overview 8
2.1 Target Variables 8
2.2 Explanatory Variables 8
2.3 Additional variables 8
2.4 Data Quality Snapshot 9
3. Analytical Findings 10
3.1 Employment Structure — Cross-Country Exposure in NACE H 10
3.1.1 Objective 10
3.1.2 Structural Concentration 10
3.1.3 Temporal Dynamics 11
3.1.4 Gender Composition 11
3.1.5 Analytical Implication 12
3.2 Injury Incidence — Cross-Country Comparison (NACE H) 13
3.2.1 Objective 13
3.2.2 Absolute Burden 13
3.2.3 Normalized Incidence 13
3.2.4 Temporal Stability 14
3.2.5 Gender Patterns 15
3.2.6 Severity Structure 15
3.2.7 Analytical Implication 16
3.3 Injury Incidence — Italy Cross-Sector Benchmark (A–U) 17
3.3.1 Objective 17
3.3.2 Absolute Distribution 17
3.3.3 Normalized Risk Comparison 18
3.3.4 Gender Dimension 20
3.3.5 Analytical Implication 20
3.4 Accident Circumstance Structure — Cross-Country Profile (NACE H) 21
3.4.1 Objective 21
3.4.2 Structural Distribution 21
3.4.3 Italy vs Peer Profile 22
3.4.4 Demographic Structure 23
3.4.5 Analytical Implication 24
3.5 Employment Structure — Italy Cross-Sector (NACE A–U) 25
3.5.1 Objective 25
3.5.2 Sectoral Distribution 25
3.5.3 Temporal Dynamics 26
3.5.4 Gender Composition 26
3.5.5 Analytical Implication 27
4. Synthesis of Patterns & Relationships 28
4.1 Size vs Risk: Cross-Country Structure 28

4.2 Structural Accident Mechanisms 29
4.3 Sector Position of Transportation in Italy 29
4.4 Exposure and Injury Relationship 29
5. Strategic Insights & Implications 32
5.1 Key Findings 32
5.2 Implications of Observed Patterns 32
5.3 Analytical Limitations 32
5.4 Future Analytical Extensions 33
5.5 Final Takeaway 33
6. Use of Artificial Intelligence Tools 34
7. Appendices 35
Appendix 7.1 – Supplementary Tables 35
Table A7.1.1 — Average annual employment in NACE H by major country groups and gender composition, 2014–2023 35
Table A7.1.2 — Severity of Workplace Injuries in the Transportation and Storage Sector (NACE H): Fatal and Non-Fatal Accidents, 2014–2023 36
Table A7.1.3 — Average Injury Incidence in Transportation and Storage, NACE H, Across the Top-10 European Economies, 2014–2023 37
Table A7.1.4 — Injury Incidence by Economic Activity in Italy, NACE A–U, Average 2014–2023 38
Table A7.1.5 — Distribution of Accident Circumstances in NACE H, Top-10 Countries, Average 2014–2023 40
Table A7.1.6 — Gender Composition of Employment: Transportation and Storage vs Total Economy in Italy, Average 2014–2023 43
Appendix 7.2 – Supplementary Charts 45
Figure A7.2.1 — Average Employment in NACE H, Italy vs Top Sector Economies (2014–2023) 45
Figure A7.2.2 — Average Injury Incidence (Casualties per 1,000 Employees, 2014–2023) 45
Appendix 7.3 – Reference Classification Tables 46
Table 7.3.1 – Dimension Table: Countries 46
Table 7.3.2 – Dimension Table: Economic Activities 48
Table 7.3.3 – Dimension Table: Sex 49
Table 7.3.4 – Dimension Table: Age Classes 49
Table 7.3.5 – Dimension Table: Injury Severity 50
Table 7.3.6 – Dimension Table: Circumstance Categories and Subcategories 50
Appendix 7.4 — Key Performance Indicators (KPIs) 54
Table 7.4.1 – Employment Structure — Cross-Country Context (Section 3.1) 54
Table 7.4.2 – Injury Incidence — Cross-Country Comparison (Section 3.2) 54
Table 7.4.3 – Injury Incidence — Cross-Sector Comparison in Italy (Section 3.3) 54
Table 7.4.4 – Accident Circumstance Structure (Section 3.4) 55
Table 7.4.5 – Employment Context of NACE H in Italy (Section 3.5) 55

Document Control

Version	Date	Author	Description
0.1	Dec. 28, 2025	Giulio Galbiati	Initial document structure created based on C1 examples.
0.2	Dec. 31, 2025	Giulio Galbiati	Drafted main sections; defined project scope and analytical objectives.
0.3	Feb. 26, 2026	Giulio Galbiati	Aligned formatting with the Data Curation document; defined EDA structure and key analytical components.
0.4	Mar. 1, 2026	Giulio Galbiati	Restructured analytical framework; aligned EDA with dual comparative design.
0.5	Mar. 2, 2026	Giulio Galbiati	Streamlined analytical scope and simplified demographic dimensions.
0.6	Mar. 8, 2026	Giulio Galbiati	Integrated analytical results and visualizations (Sections 3.1–3.5).
0.7	Mar. 9, 2026	Giulio Galbiati	Completed Sections 4–5 and added Appendix KPI tables and supporting material.
0.8	Mar. 10, 2026	Giulio Galbiati	Final editorial review, formatting alignment, and cross-reference verification.
1.0	Mart. 11, 2026	Giulio Galbiati	Final version submitted.

Document Status

This document presents the exploratory data analysis (EDA) performed on the curated dataset developed during the Data Curation phase of the project.

The analysis is descriptive and intended to uncover patterns, trends, and relationships that inform the defined business problem.

Exploratory Data Analysis (EDA)

Project: Work-related injuries in the Italian transportation and storage sector

Author: Giulio Galbiati

Date: March 10, 2026

Version: 1.0

1. Executive Overview

1.1 Purpose

This document presents the exploratory data analysis (EDA) of work-related injuries in the Italian Transportation and Storage sector (NACE Rev. 2 – Activity H), based on curated EUROSTAT datasets covering the period 2014–2023.

The analysis examines employment-adjusted injury incidence in order to identify temporal patterns, structural risk drivers, cross-country differences, and the sector's relative positioning within the Italian economy.

The analysis builds upon the curated relational dataset developed during the Data Curation phase and focuses on normalized indicators expressed as casualties per 1,000 employees.

1.2 Analytical Roadmap

The analysis proceeds in four stages.

- First, workforce exposure in the Transportation and Storage sector is examined across European countries to establish the structural distribution of employment.
- Second, employment-adjusted injury incidence is compared across countries in order to assess relative risk.
- Third, the sector's risk profile is benchmarked against other economic activities within the Italian economy.
- Finally, accident circumstances are analyzed to identify operational contexts most frequently associated with injuries.

1.3 Key Questions

- How is workforce exposure distributed across European countries in the Transportation and Storage sector (NACE H)?
- How does employment-adjusted injury incidence vary across peer European countries, and where does Italy position within this comparison?
- How does the Transportation and Storage sector compare with other economic sectors in Italy in terms of employment-adjusted injury risk?
- Which accident circumstances appear most frequently associated with injuries in the sector?
- To what extent do workforce size and injury incidence appear structurally related across countries and sectors?

1.4 Headline Findings

- Key Temporal Pattern: Injury incidence in the Transportation and Storage sector remained broadly stable over the 2014–2023 period, with moderate fluctuations but no persistent upward or downward long-term trend.

- Cross-Country Risk Positioning: Italy's injury incidence in NACE H is broadly aligned with the European peer group, indicating comparable occupational risk levels relative to other major sector economies.
- Dominant Accident Mechanisms: Work-related injuries are strongly concentrated in a limited set of operational circumstances associated with transport operations, manual handling, and vehicle-related activities.
- Sector Benchmark within the Italian Economy: Within Italy's economic structure, Transportation and Storage exhibits a higher injury incidence than the national economy average, confirming its classification as a structurally higher-risk sector.
- Primary Analytical Insight: Most cross-country differences in absolute accident counts are explained by workforce size, making employment-normalized indicators essential for meaningful safety comparisons.

1.5 Analytical Positioning

This analysis is descriptive and comparative in nature. Injury incidence is evaluated using employment-adjusted indicators (casualties per 1,000 employees) in order to ensure comparability across countries and sectors.

The objective is not to establish causal relationships, but to identify statistically grounded patterns that may inform occupational safety strategies and guide future analytical investigation.

2. Data & Variables Overview

This section summarizes the main variables used in the analysis and provides a brief overview of the dataset structure and data-quality considerations.

2.1 Target Variables

The analysis focuses on indicators describing the scale and intensity of work-related injuries in the Transportation and Storage sector.

- Employees – number of employed persons used as the exposure denominator.
- Casualties per 1,000 employees – normalized injury incidence indicator used for cross-country and cross-sector comparison.
- Total casualties – absolute number of recorded work-related injuries.
- Fatal vs non-fatal ratio – relative proportion of fatal accidents within total recorded incidents.

These variables capture both the absolute burden of injuries and the relative occupational risk after adjusting for workforce size.

2.2 Explanatory Variables

The analysis examines how injury patterns vary across several contextual dimensions:

- Country – geographic unit used for cross-country comparison within NACE H.
- Year – temporal dimension covering the 2014–2023 observation window.
- Circumstance category – ESAW classification describing the operational context of the accident.
- Gender – demographic dimension used to assess differences in injury incidence between male and female workers.
- Economic activity (Italy A–U) – sector classification used for within-country comparison across the Italian economy.

2.3 Additional variables

The curated dataset includes additional attributes that are not central to the main analysis but remain available for descriptive exploration.

- Age class – demographic categorization of workers involved in accidents.
- Severity classification – distinction between fatal and non-fatal accidents.

These variables are retained in the database structure but are not used as primary analytical dimensions in the core EDA.

2.4 Data Quality Snapshot

The dataset integrates multiple EUROSTAT statistical sources and reflects several structural characteristics:

- Time coverage: 2014–2023, providing a consistent ten-year observation window.
- Handling of missing values: structural gaps in national reporting were identified and treated explicitly, with estimated values used only for aggregate descriptive analysis.
- Normalization: injury counts are expressed relative to employment levels (casualties per 1,000 employees) to ensure meaningful comparison across countries and sectors with different workforce sizes.

3. Analytical Findings

3.1 Employment Structure — Cross-Country Exposure in NACE H

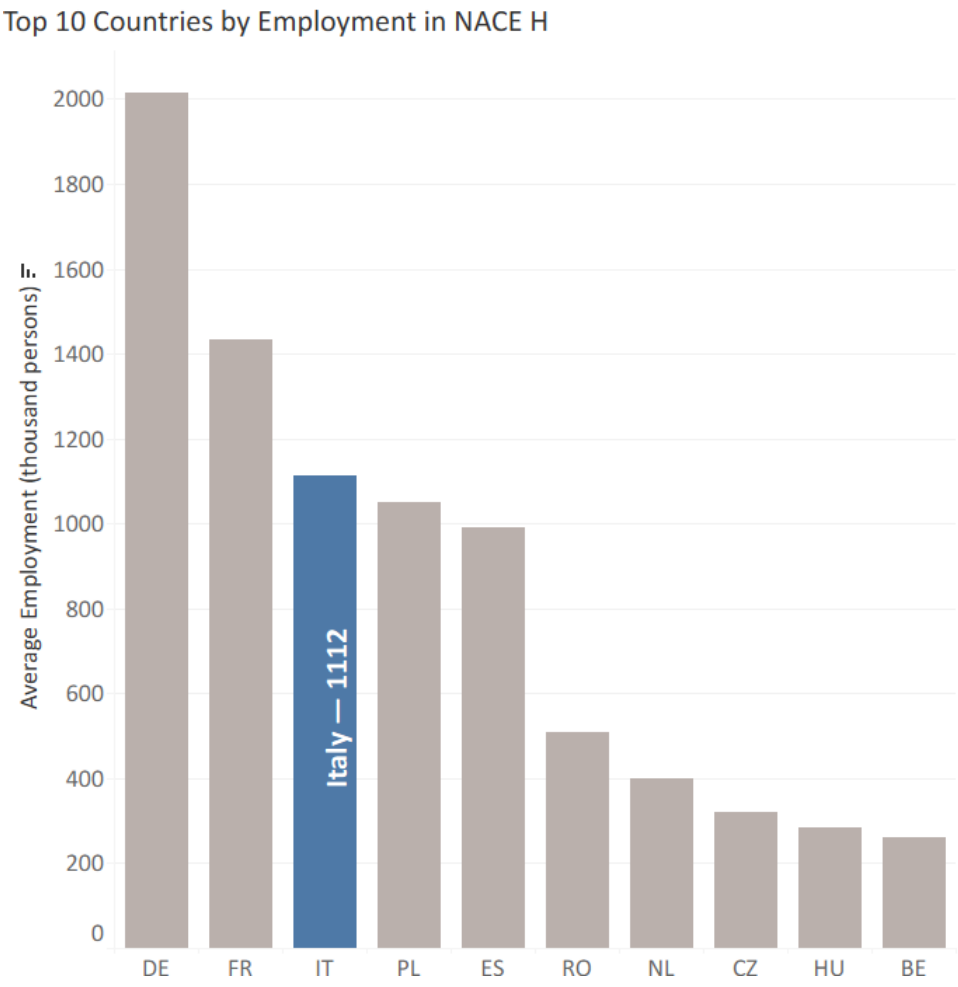
3.1.1 Objective

Before analyzing injury outcomes, it is necessary to understand the distribution of employment exposure in the transportation and storage sector (NACE H). Differences in workforce size across countries can influence the number of recorded injuries. This section therefore examines the cross-country distribution of employment, its evolution over 2014–2023, and its gender composition.

3.1.2 Structural Concentration

Employment in NACE H is concentrated in a limited number of large European economies. When averaged over the period 2014–2023, the top ten countries account for the majority of total sector employment, reflecting differences in economic scale and logistics activity.

Figure 3.1 — Employment concentration in NACE H: Top 10 countries by average employment (2014–2023).

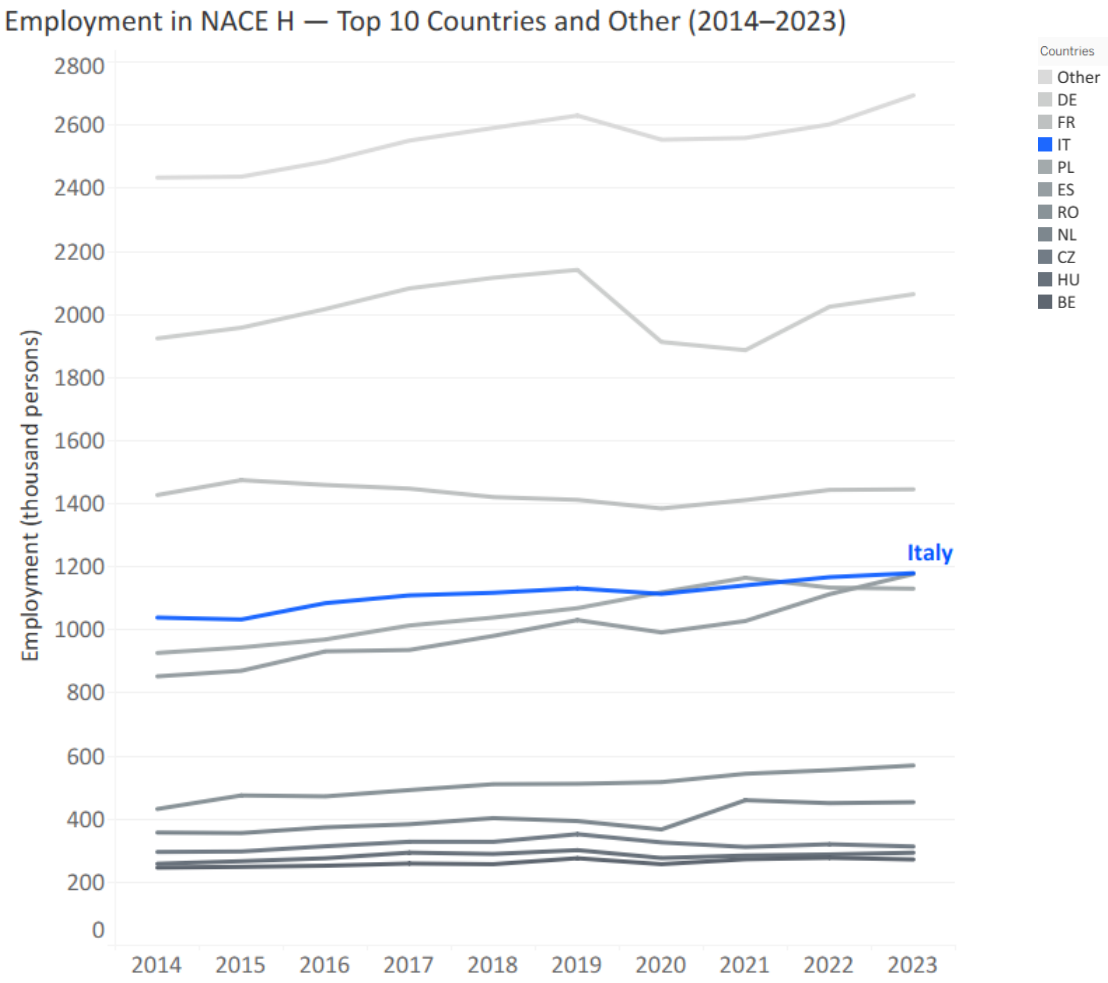


Source: Eurostat — Labour Force Survey (LFS); author's calculations.

3.1.3 Temporal Dynamics

Total employment in the sector remained broadly stable over the decade, with moderate year-to-year variation. This stability suggests that changes in injury indicators are unlikely to be driven by large shifts in workforce size.

Figure 3.2 — Employment trends in NACE H across the ten largest sector economies and other EU countries, 2014–2023.

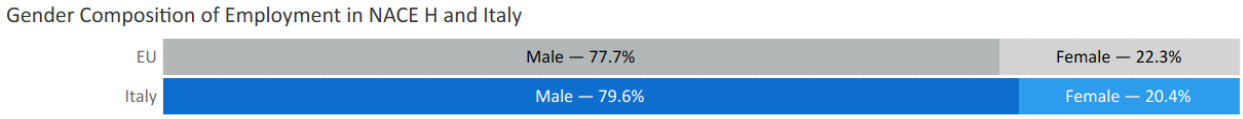


Source: Eurostat employment statistics, author's calculations.

3.1.4 Gender Composition

The sector workforce is predominantly male, with female workers representing a smaller share of total employment throughout the period. This pattern reflects the occupational structure of transportation and logistics activities.

Figure 3.3 — Gender composition of employment in NACE H across EU countries, 2014–2023.



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

3.1.5 Analytical Implication

These structural characteristics provide important context for the injury analysis.

Employment exposure is geographically concentrated, relatively stable over time, and predominantly male, reinforcing the need to interpret injury outcomes using exposure-adjusted indicators in the following sections.

Monitoring employment exposure and demographic composition is therefore essential for correctly interpreting cross-country injury comparisons.

A set of derived indicators summarizing employment exposure in NACE H was computed to support interpretation of injury rates.

These include employment concentration across countries, sectoral employment growth, and gender composition metrics.

The full indicator definitions and computed values are reported in Appendix 7.4.

The next section evaluates cross-country differences in injury incidence in the transportation and storage sector, using exposure-adjusted indicators.

3.2 Injury Incidence — Cross-Country Comparison (NACE H)

3.2.1 Objective

This section compares work-related injury incidence across the Top-10 European countries in NACE H (2014–2023), with specific focus on Italy's positioning.

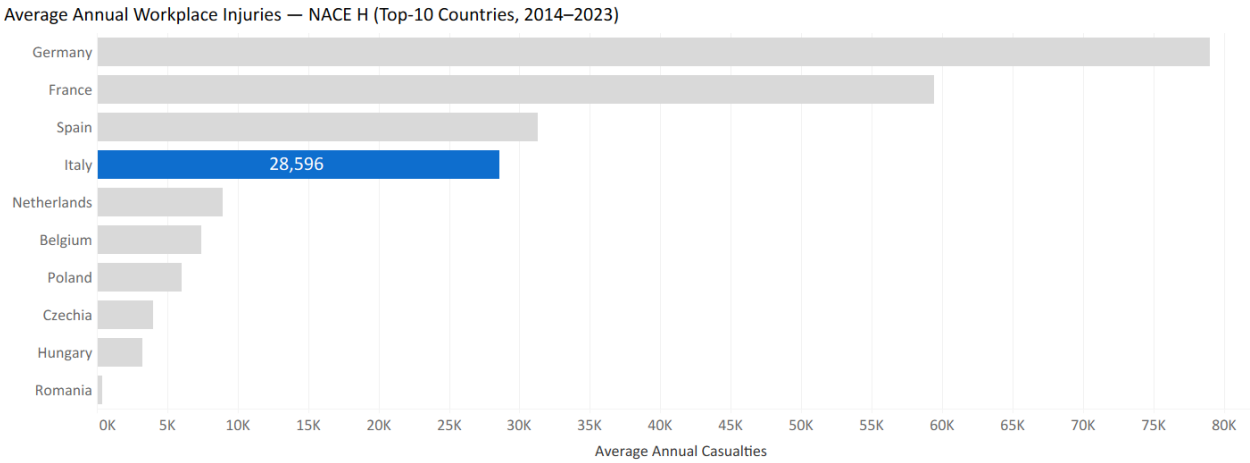
Two analytical layers are considered: volume (absolute counts) and risk intensity (normalized rates).

3.2.2 Absolute Burden

Absolute casualty counts vary markedly across countries and primarily reflect the scale of sector employment.

They are useful for describing burden, but they do not indicate relative occupational risk without normalization.

Figure 3.4 — Average Annual Workplace Injuries — NACE H (Top-10 EU Countries, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

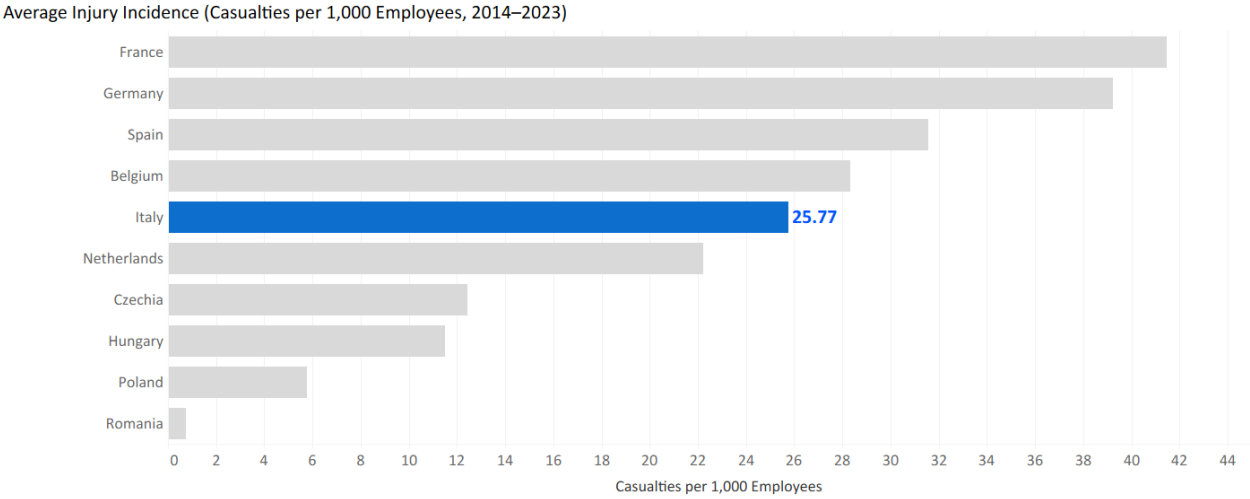
3.2.3 Normalized Incidence

When normalized by employment, cross-country rankings change significantly compared with absolute injury totals.

Several countries with large employment bases show moderate risk levels, while others exhibit higher injury incidence relative to workforce size.

Italy moves from a mid-range position in absolute counts to a comparatively lower-risk position among the peer group, highlighting the importance of employment-adjusted indicators for meaningful cross-country comparison.

Figure 3.5 — Average Injury Incidence — Casualties per 1,000 Employees (2014–2023)



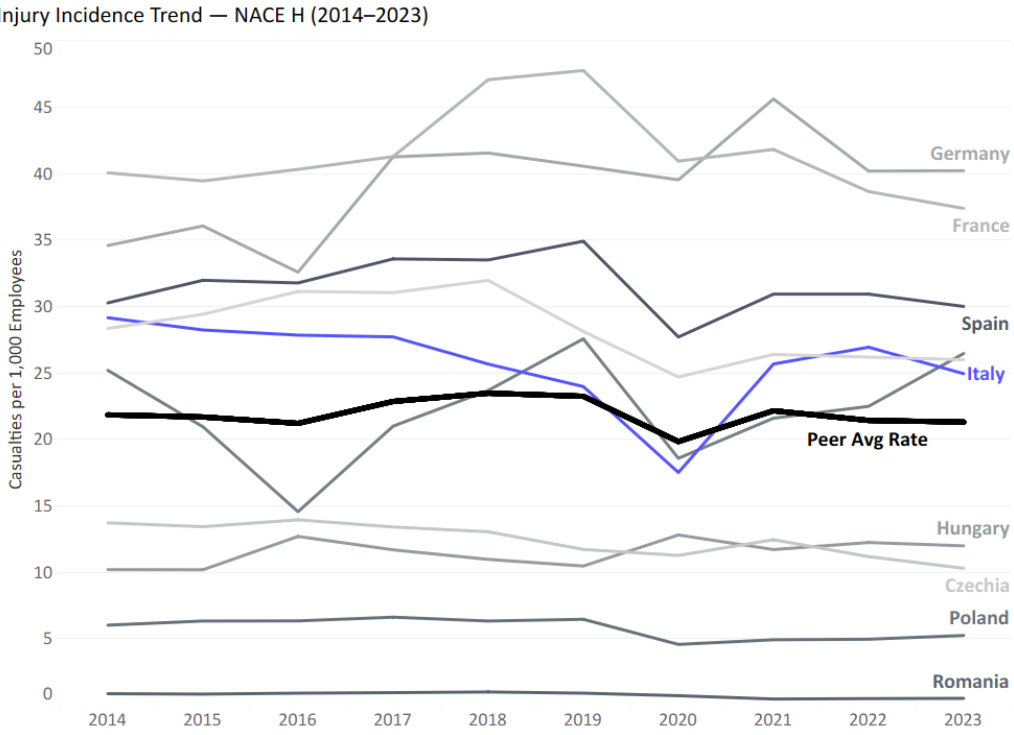
Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

3.2.4 Temporal Stability

While injury rates fluctuate across individual countries, the peer-average trend remains relatively stable over the observation period.

This suggests that short-term variations largely reflect country-specific fluctuations rather than a structural change in sectoral risk.

Figure 3.6 — Injury Incidence Trend — NACE H (2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

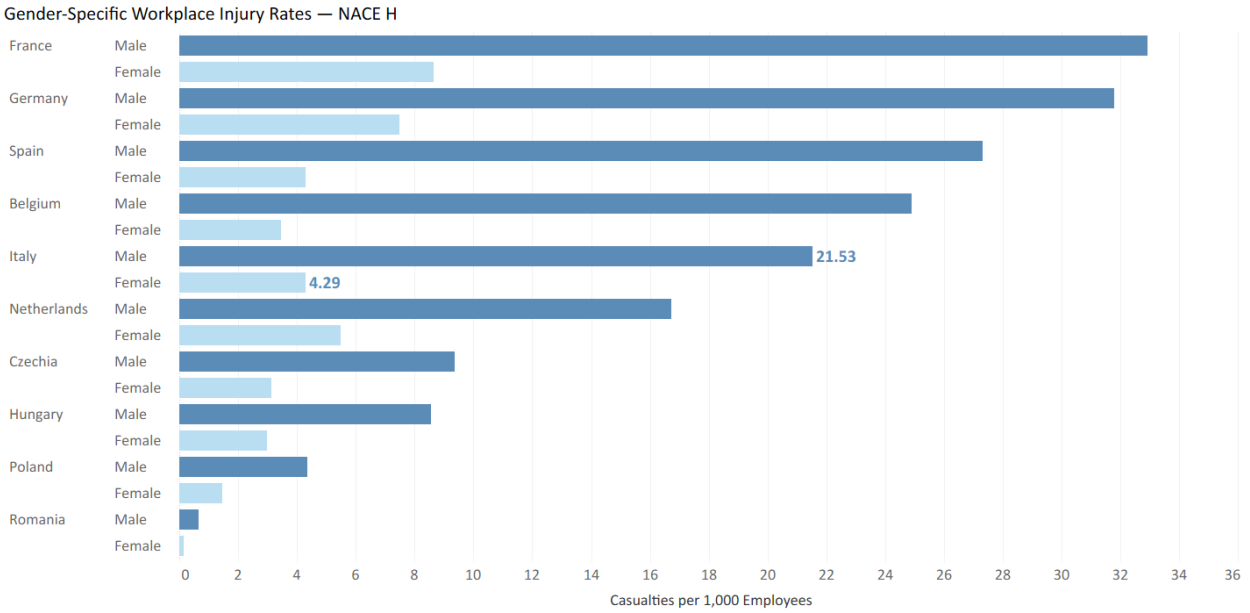
3.2.5 Gender Patterns

Across all countries, injury incidence among male employees substantially exceeds that observed among female employees.

This pattern reflects the higher concentration of men in operational and physically demanding roles within the transportation and storage sector.

Italy's male and female injury incidence levels are broadly aligned with the middle of the peer-country distribution, while the male rate remains substantially above the female rate across all countries.

Figure 3.7 — Gender-Specific Injury Incidence — Casualties per 1,000 Employees (NACE H, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

3.2.6 Severity Structure

Fatal accidents represent only a very small fraction of total incidents within the dataset. Over the 2014–2023 period, 4,267 fatalities were recorded out of approximately 2.28 million reported workplace injuries, corresponding to an average fatal share of about 0.19%.

The annual fatal share remains relatively stable throughout the period, fluctuating within a narrow range between 0.16% and 0.22%.

Interpretation remains cautious because the available severity classification distinguishes only between fatal accidents and non-fatal accidents involving four or more days of absence, limiting the granularity of severity analysis.

Annual data on fatal and non-fatal accidents are provided in Appendix A7.1.

3.2.7 Analytical Implication

Cross-country analysis highlights three key insights:

- Injury burden and injury risk are distinct phenomena.
- Absolute casualty counts primarily reflect workforce size, whereas normalized indicators capture underlying occupational risk.
- Italy's structural exposure does not automatically imply higher risk intensity.
- Although Italy ranks among the larger countries in terms of total injuries, its employment-adjusted incidence places it closer to the lower range of the peer group.
- Normalized indicators are essential for meaningful comparison.
- Without employment adjustment, cross-country comparisons may misrepresent relative risk levels.

A small set of key performance indicators summarizes the main findings of the cross-country comparison presented in this section.

The indicators are based on employment-adjusted injury incidence rates, expressed as casualties per 1,000 employees, allowing comparison across countries with different workforce sizes.

The complete KPI table is provided in Appendix A7.4, where average values for the period 2014–2023 are reported for the peer-country group and for Italy.

The next section examines injury incidence within Italy's economic structure, benchmarking the Transportation and Storage sector (NACE H) against all major economic activities (A–U).

3.3 Injury Incidence — Italy Cross-Sector Benchmark (A–U)

3.3.1 Objective

While the previous section examined injury incidence across countries within the Transportation and Storage sector (NACE H), this section shifts the analytical perspective to Italy's internal economic structure.

The objective is to benchmark the injury incidence of the Transportation and Storage sector against the full set of Italian economic activities (NACE A–U) over the period 2014–2023.

Two complementary dimensions are considered:

- Absolute injury burden, reflecting the total number of casualties recorded in each sector.
- Normalized injury incidence, expressed as casualties per 1,000 employees, which enables meaningful comparison across sectors with different workforce sizes.

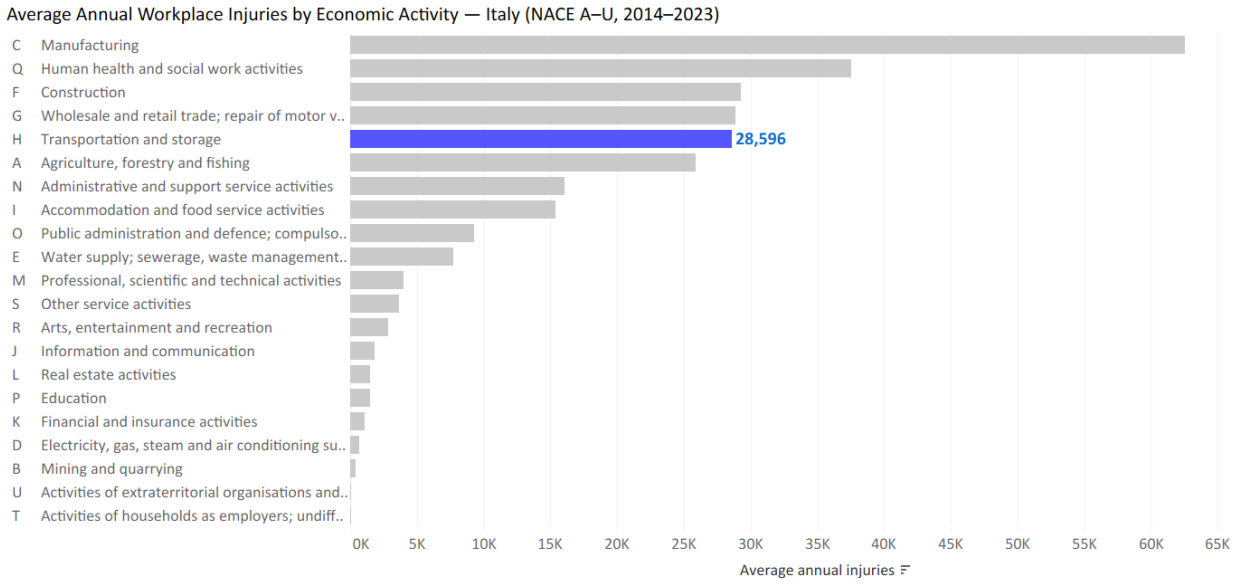
This comparison allows the Transportation and Storage sector to be positioned within the broader Italian occupational risk landscape, identifying whether the sector exhibits above-average, average, or below-average injury intensity relative to the national economic structure.

3.3.2 Absolute Distribution

Casualty totals vary substantially across economic activities and largely reflect differences in workforce size.

Larger sectors therefore record higher numbers of injuries, making absolute counts an indicator of exposure rather than risk intensity. Exposure-adjusted indicators are required for meaningful cross-sector comparison.

Figure 3.8 — Average Annual Workplace Injuries by Economic Activity — Italy (NACE A–U, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

Workplace injuries are unevenly distributed across economic activities, with larger sectors accounting for a higher share of total casualties.

Absolute counts therefore reflect employment scale and should be interpreted alongside normalized indicators.

To assess relative risk across sectors, the following section examines injury incidence normalized by employment.

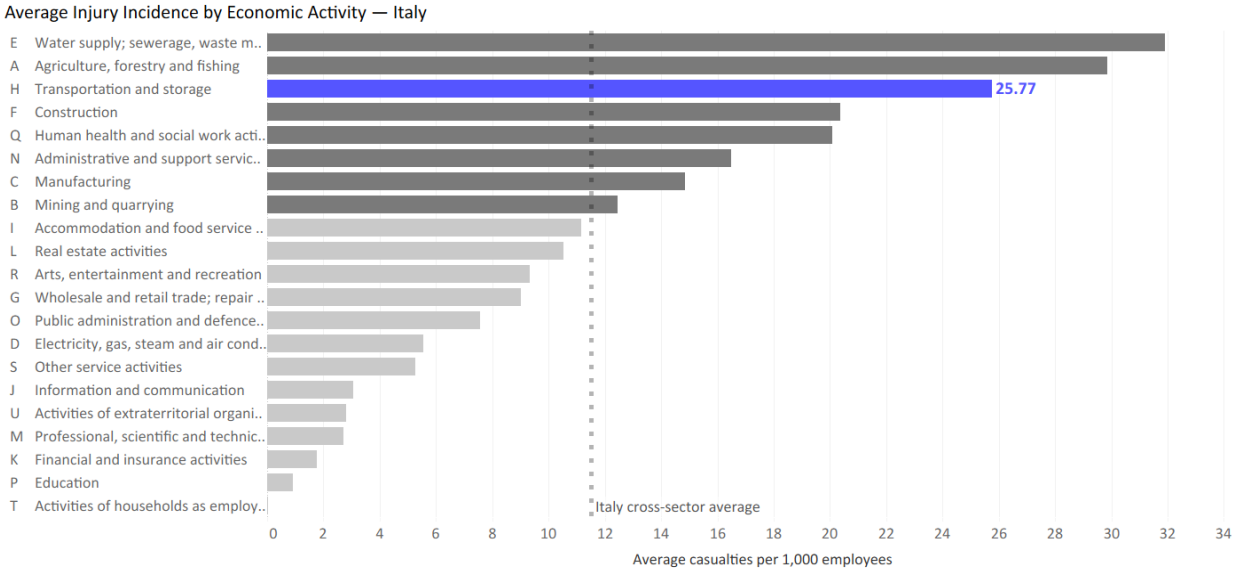
3.3.3 Normalized Risk Comparison

To assess relative occupational risk across sectors, injury incidence must be evaluated relative to workforce size.

This section therefore examines workplace injury rates expressed as casualties per 1,000 employees across Italian economic activities (NACE A–U) over the period 2014–2023.

By normalizing casualty counts by employment, the analysis enables meaningful comparison of sectoral risk intensity independent of differences in workforce scale.

Figure 3.9 — Average Injury Incidence by Economic Activity — Italy (Casualties per 1,000 Employees, NACE A–U, 2014–2023)

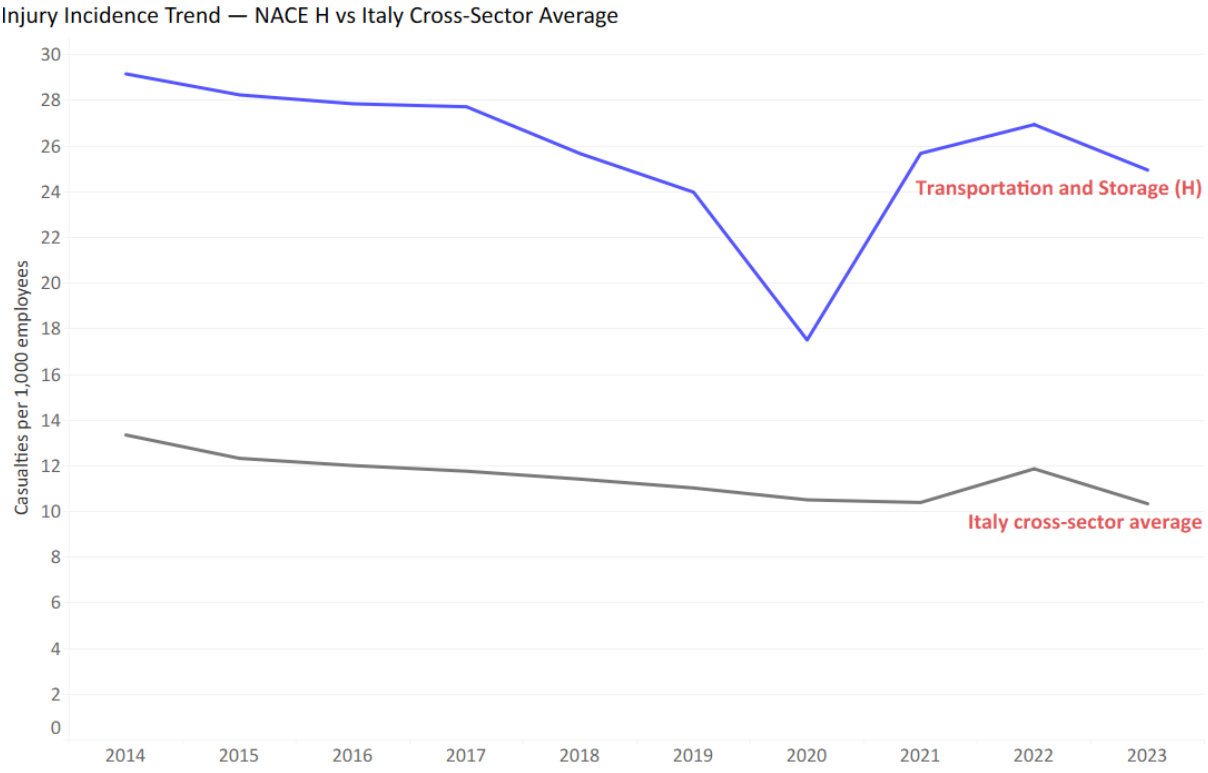


Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

Sectoral injury incidence varies markedly across the Italian economy.

Transportation and Storage (NACE H) appears among the higher-risk activities, though it is not the sector with the highest injury rate. The normalized comparison therefore provides a more accurate perspective on relative occupational risk.

Figure 3.10 — Injury Incidence Trend: NACE H vs Cross-Sector Average — Italy (Casualties per 1,000 Employees, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The trend comparison shows that injury incidence in Transportation and Storage (NACE H) consistently remains above the cross-sector average in Italy over the period 2014–2023.

While annual fluctuations occur, the relative position of the sector within the national risk distribution remains broadly stable.

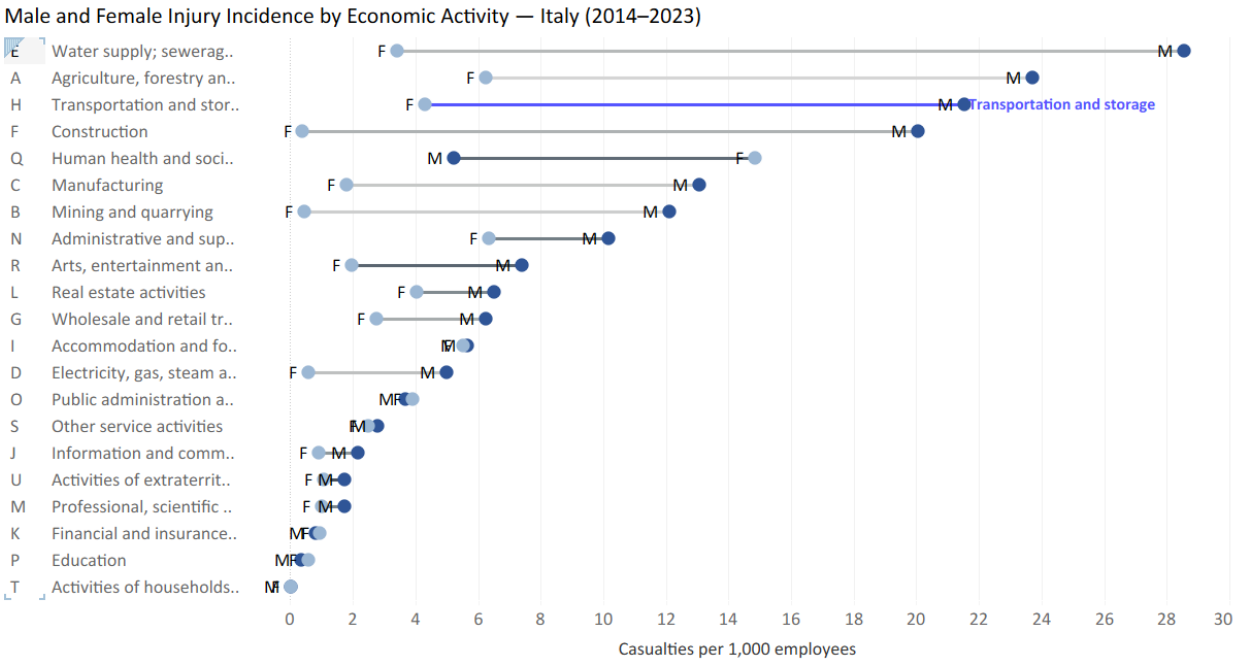
3.3.4 Gender Dimension

Gender-differentiated injury rates reveal a clear disparity between male and female workers across economic activities.

Male injury incidence exceeds female incidence across all sectors of the Italian economy.

Transportation and Storage (NACE H) shows one of the larger gender gaps, reflecting the sector's occupational structure and workforce composition.

Figure 3.11 — Injury Incidence by Gender across Economic Activities — Italy (Casualties per 1,000 Employees, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

3.3.5 Analytical Implication

Within Italy, the analysis highlights several structural patterns.

First, injury risk varies materially across economic activities, confirming the importance of sector-specific exposure conditions.

Second, Transportation and Storage emerges as a sector with consistently above-average injury incidence, though not among the most extreme risk environments in the Italian economy.

Third, cross-sector benchmarking underscores the value of sector-focused safety analysis when interpreting occupational risk patterns.

Key summary indicators (KPIs) for the cross-sector comparison are reported in Appendix 7.4.

The next section shifts from sectoral comparison to the structure of accident circumstances within the Transportation and Storage sector, examining cross-country patterns across the Top-10 European economies.

3.4 Accident Circumstance Structure — Cross-Country Profile (NACE H)

3.4.1 Objective

This section examines the distribution of accident circumstances in the Transportation and Storage sector (NACE Rev. 2 – Activity H) across the Top-10 European countries, with specific reference to Italy.

Because ESAW circumstance classifications describe contextual attributes of accidents rather than mutually exclusive event counts, the analysis focuses on relative distributional shares of circumstance subcategories rather than additive totals.

The objective is to identify the dominant accident mechanisms characterizing the sector and to assess whether Italy's accident profile diverges from the peer-country structure.

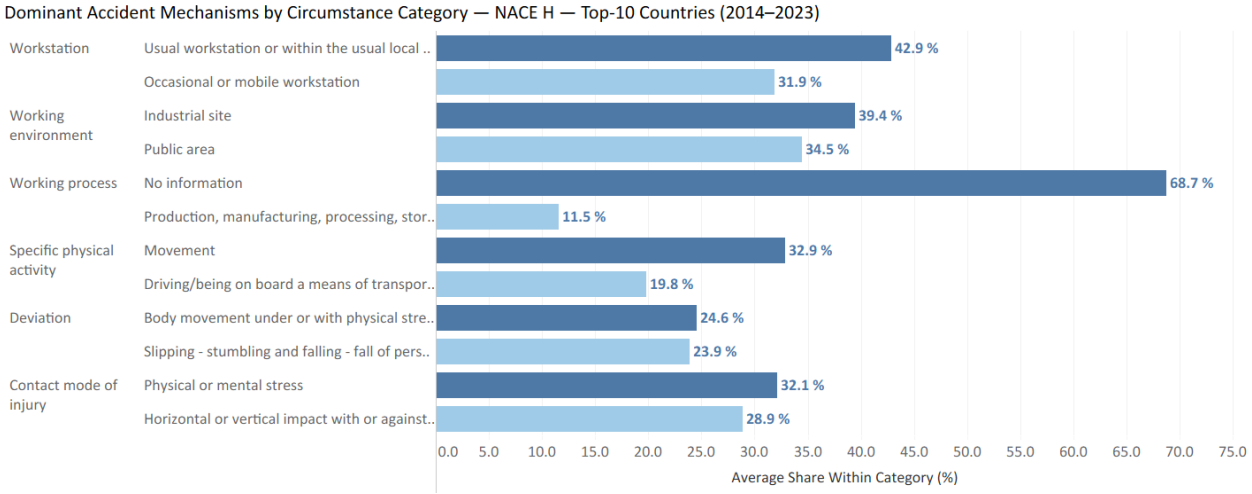
3.4.2 Structural Distribution

Across the ESAW circumstance taxonomy, a limited number of subcategories account for the majority of observed incidents.

This concentration is observed consistently across countries and remains broadly stable throughout the 2014–2023 period, indicating the presence of persistent structural accident mechanisms rather than short-term anomalies.

The distribution therefore reflects the operational contexts in which workplace injuries most frequently occur within the Transportation and Storage sector.

Figure 3.12 — Dominant Accident Mechanisms by Circumstance Category — NACE H — Top-10 Countries (Average Share Within Category, 2014–2023)



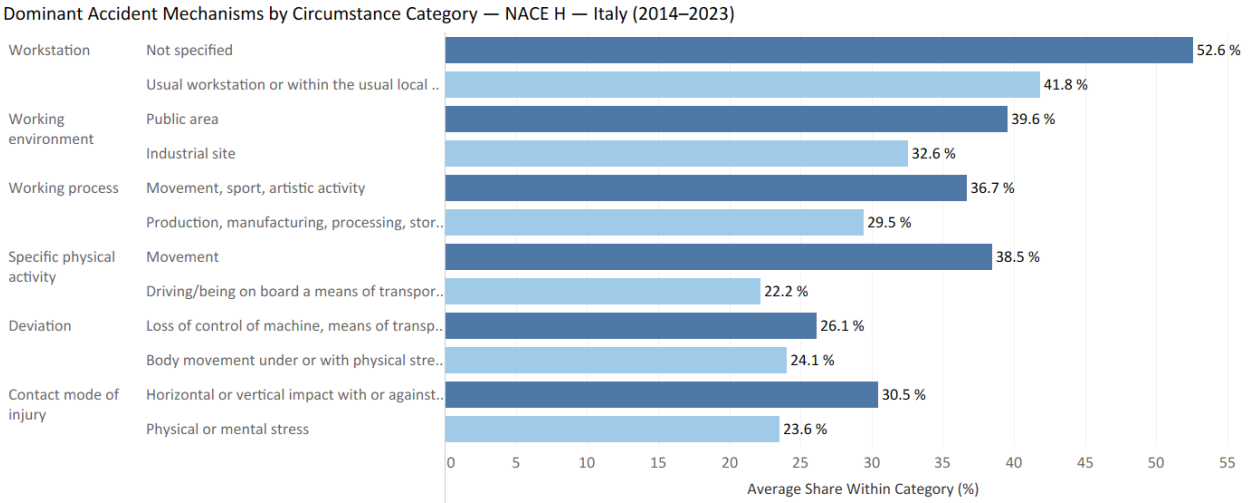
Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The figure displays the two most frequent subcategories within each ESAW circumstance category, measured as the average share within category across the Top-10 countries during the 2014–2023 period.

3.4.3 Italy vs Peer Profile

To assess whether the Italian Transportation and Storage sector exhibits a distinct accident structure, the distribution of accident circumstances is examined for Italy alone.

Figure 3.13 — Dominant Accident Mechanisms by Circumstance Category — NACE H — Italy (Average Share Within Category, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The Italian distribution shows a broadly similar structure to that observed across the Top-10 countries.

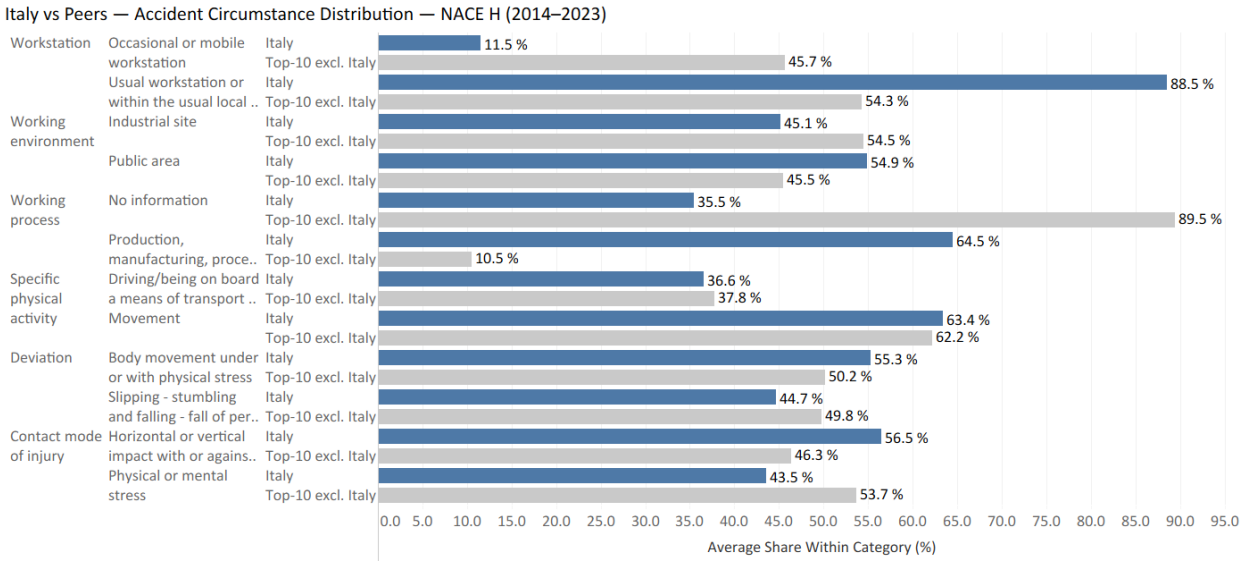
Across most circumstance categories, the dominant accident mechanisms reflect operational contexts typical of transport and logistics activities, such as worker movement, transport operations, and impact-related injuries.

Some differences emerge in the ranking of specific subcategories, particularly within the Workstation and Working process dimensions, partly reflecting variations in reporting practices and classification coverage.

However, these variations do not indicate a fundamentally different accident structure.

Overall, the mechanisms underlying workplace accidents in the Italian Transportation and Storage sector remain largely consistent with those observed across comparable European economies.

Figure 3.14 — Italy vs Peer Countries: Distribution of Dominant Accident Mechanisms — NACE H (Average Share Within Category, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The figure compares Italy with the aggregate of peer countries (Top-10 excluding Italy) for the dominant accident mechanisms identified in the previous analysis. Values represent the average share within category across the 2014–2023 period, allowing direct comparison of the relative importance of each mechanism.

Overall, the comparison confirms that Italy's accident circumstance structure remains broadly aligned with the peer-country pattern, with only moderate differences observed in selected subcategories.

3.4.4 Demographic Structure

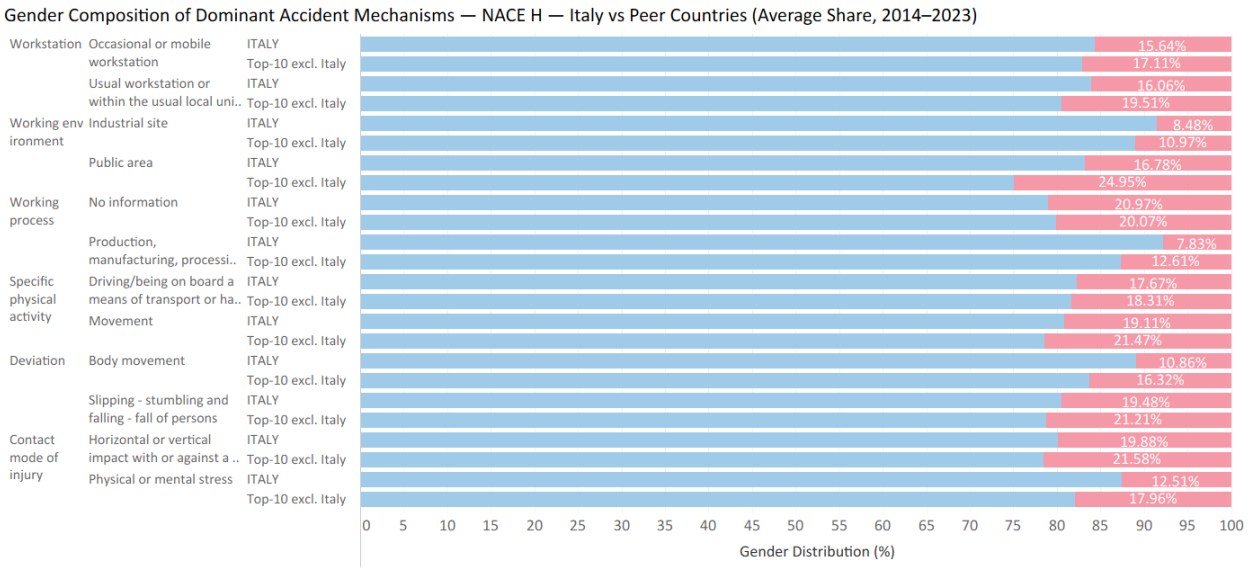
Accidents in the Transportation and Storage sector predominantly involve male workers, reflecting the strongly male-skewed composition of employment in this economic activity.

Across all dominant accident mechanisms identified in the previous analysis, male workers consistently account for the large majority of reported cases, while female shares remain comparatively limited.

The comparison between Italy and peer countries shows broadly similar gender distributions, suggesting that accident patterns largely mirror underlying workforce exposure rather than indicating mechanism-specific gender effects.

Age-based comparisons are not conducted due to incompatibilities between ESAW injury age classifications and the broader age groupings available in employment datasets, which prevent the construction of age-standardized indicators.

Figure 3.15 — Gender Composition of Dominant Accident Mechanisms — NACE H — Italy vs Peer Countries (Average Share, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The figure compares the gender distribution of casualties within the dominant accident mechanisms identified for the sector.

Values represent the average share of male and female casualties within each mechanism over the 2014–2023 period for Italy and the aggregate of peer countries (Top-10 excluding Italy).

3.4.5 Analytical Implication

Circumstance analysis highlights three key structural insights:

- Accident mechanisms in the Transportation and Storage sector are concentrated within a limited set of operational contexts.
- Italy's accident profile closely aligns with patterns observed across peer countries, indicating broadly comparable operational risk structures.
- Sectoral injury risk appears to be driven primarily by structural characteristics of transport and logistics activities rather than by isolated national anomalies.

The following section examines the employment structure across economic activities within Italy (NACE Rev. 2 – Activities A–U), providing the national sectoral context in which Transportation and Storage (Activity H) operates.

3.5 Employment Structure — Italy Cross-Sector (NACE A–U)

3.5.1 Objective

This section examines the employment structure of the Italian economy across NACE Rev. 2 activities (A–U) over the period 2014–2023.

The objective is to position Transportation and Storage (Activity H) within the broader national sectoral landscape.

Understanding how employment is distributed across economic activities provides essential context for interpreting sectoral injury incidence discussed in Section 3.3.

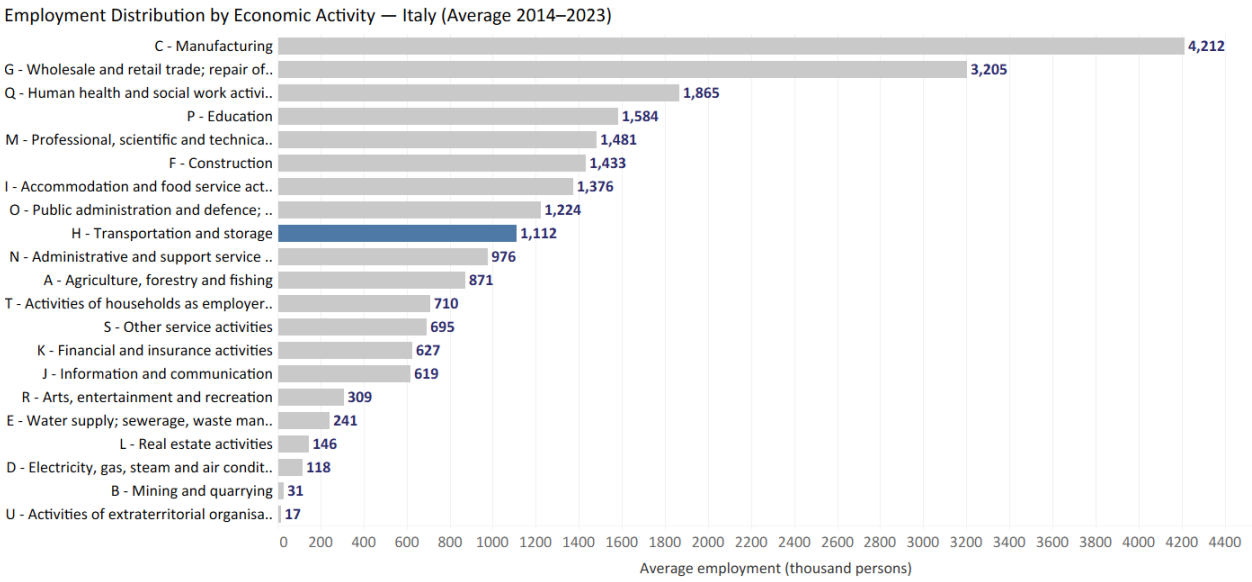
3.5.2 Sectoral Distribution

Italian economic activities are ranked according to their 10-year average employment levels.

Transportation and Storage represents a substantial share of national employment, although it does not belong to the largest sectors of the Italian economy.

Its position in the employment ranking indicates that the sector is economically relevant but not structurally dominant, making it suitable for cross-sector benchmarking of occupational injury incidence.

Figure 3.16 — Employment Distribution by Economic Activity in Italy (Average 2014–2023)



Source: Eurostat — Labour Force Survey (LFS); author's calculations.

Employment values represent the 10-year average number of employed persons in each NACE Rev. 2 activity (2014–2023). Transportation and Storage (Activity H) is highlighted to indicate its relative position within the Italian economic structure.

Employment in Italy is concentrated in a limited number of service and industrial sectors, particularly Wholesale and Retail Trade, Manufacturing, and Health and Social Work, which account for the largest shares of the workforce.

Transportation and Storage occupies a mid-ranking position in the national employment structure, confirming its relevance within the economy while remaining smaller than the largest employment sectors.

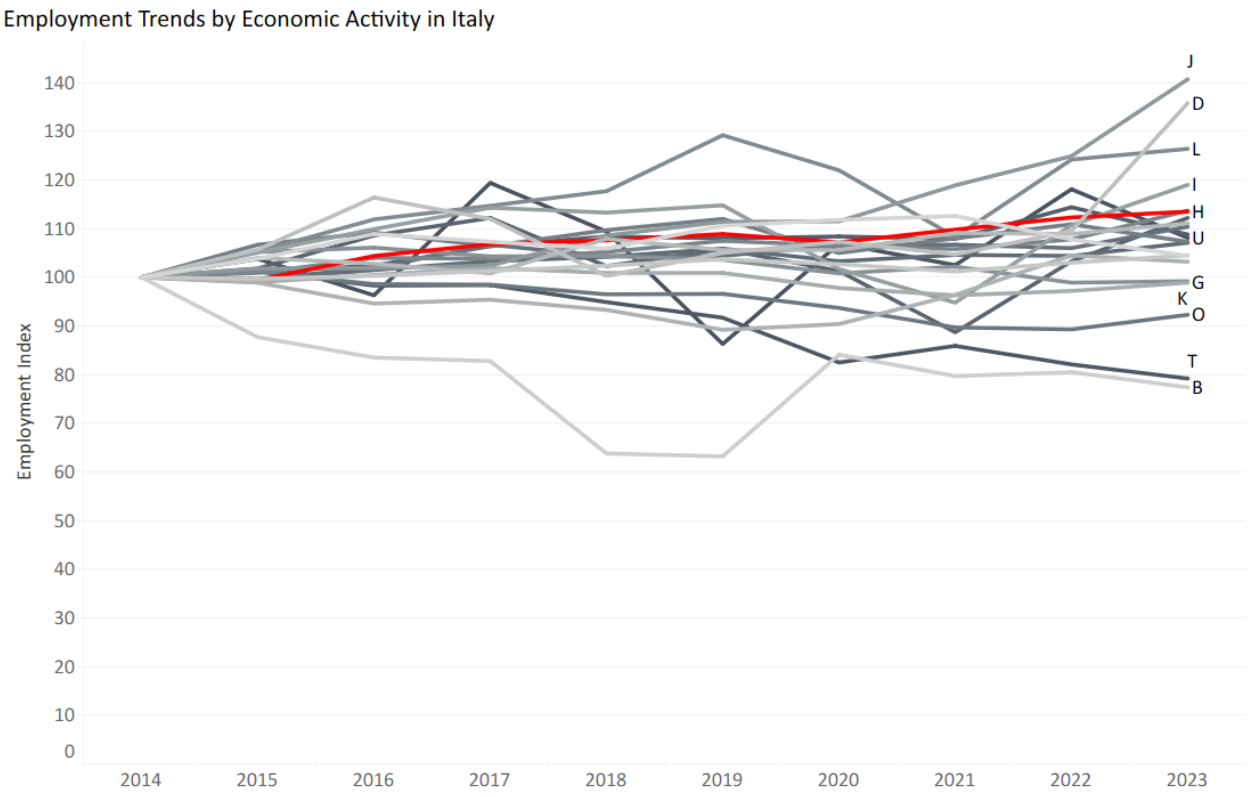
This positioning provides an appropriate context for interpreting sectoral injury incidence relative to the broader Italian economy.

3.5.3 Temporal Dynamics

Employment levels across Italian economic activities show moderate variation during the study window, with no evidence of structural discontinuities in the overall employment series.

Transportation and Storage (NACE H) exhibits relatively stable employment dynamics compared with more cyclical sectors, indicating broadly consistent exposure conditions over the observation period.

Figure 3.17 — Employment Trends by Economic Activity in Italy (Index 2014 = 100)



Source: Eurostat — Labour Force Survey (LFS); author's calculations.

Employment values are indexed to 2014 = 100 to facilitate comparison of relative changes across NACE Rev. 2 activities over the period 2014–2023.

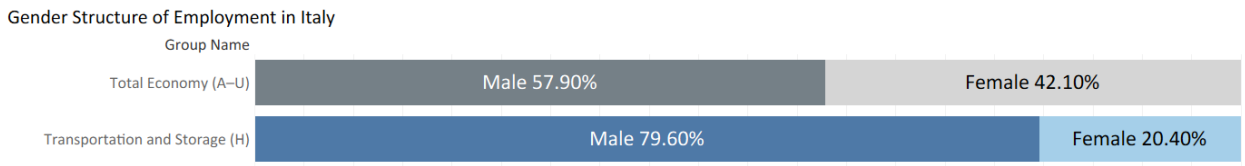
3.5.4 Gender Composition

The Italian Transportation and Storage sector is predominantly male.

Compared with the aggregate Italian workforce (NACE A–U), Activity H shows a higher concentration of male employment, indicating a structurally gender-skewed exposure profile.

This characteristic is relevant when interpreting gender-differentiated injury incidence.

Figure 3.18 — Gender Composition of Employment: Transportation and Storage vs Total Economy in Italy (Average 2014–2023)



Source: Eurostat — Labour Force Survey (LFS); author's calculations.

Employment shares represent the average distribution of male and female employment over 2014–2023. Total Economy refers to the aggregate of Italian economic activities A–U.

3.5.5 Analytical Implication

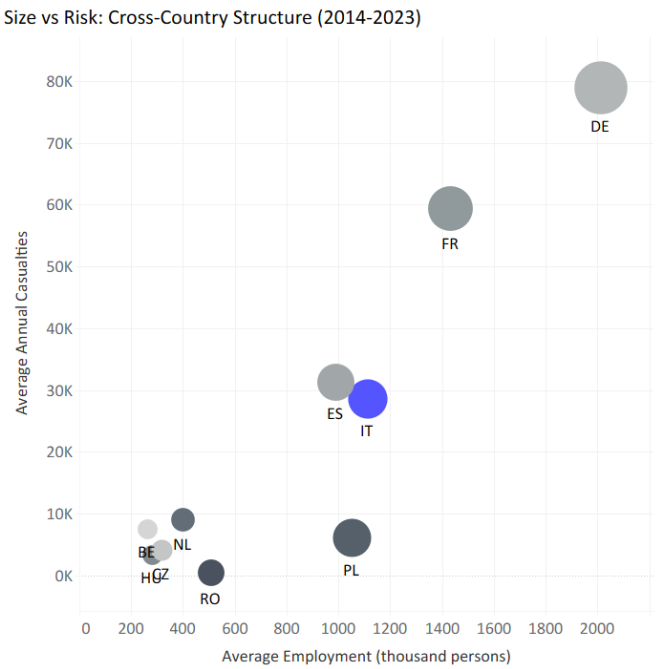
The sectoral employment analysis confirms three structural observations:

- Transportation and Storage (NACE H) represents a significant but not dominant component of the Italian economic structure.
- The sector exhibits a marked gender imbalance, which may influence the composition of observed injury patterns.
- Interpreting the cross-sector injury rates presented in Section 3.3 therefore requires consideration of the underlying employment structure.

4. Synthesis of Patterns & Relationships

4.1 Size vs Risk: Cross-Country Structure

Figure 3.19 — Employment Exposure and Injury Counts in Transportation and Storage (Top-10 EU Countries, Average 2014–2023



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

The relationship between employment size and accident counts across the ten largest European Transportation and Storage sectors shows a strong positive association. Countries with larger workforces tend to report higher numbers of accidents in absolute terms. The calculated **Pearson correlation coefficient of 0.91** confirms that casualty counts scale closely with sector size.

This pattern indicates that a substantial share of cross-country variation in absolute accident totals reflects exposure effects rather than intrinsic safety differences. Larger sectors naturally generate more incidents simply because more workers are exposed to operational risks.

However, the dispersion around the trend line suggests that sector size does not fully explain national accident levels. Countries with comparable employment volumes (e.g., Spain and Italy) display similar accident counts, while others with smaller workforces show proportionally lower totals.

These findings reinforce the methodological choice adopted in this study to rely on employment-normalized indicators (casualties per 1,000 employees) when comparing occupational risk across countries. Absolute counts primarily reflect workforce scale, whereas normalized measures provide a more meaningful assessment of relative safety performance.

4.2 Structural Accident Mechanisms

Accident circumstance analysis indicates that injury events in the Transportation and Storage sector are highly concentrated in a limited number of operational mechanisms.

Across countries, the same categories consistently dominate accident distributions, suggesting that sector-specific operational characteristics, such as vehicle operation, cargo handling, and movement within logistics environments, represent the primary structural drivers of occupational risk.

This pattern reinforces the interpretation that risk arises from intrinsic features of transport and logistics activities rather than country-specific anomalies.

4.3 Sector Position of Transportation in Italy

The cross-sector comparison presented in Section 3.3 provides the national context for interpreting the safety profile of Transportation and Storage (NACE H).

While several sectors account for larger absolute numbers of accidents due to their employment scale, normalization by workforce size shows that Transportation and Storage maintains a consistently higher injury incidence than the Italian cross-sector average.

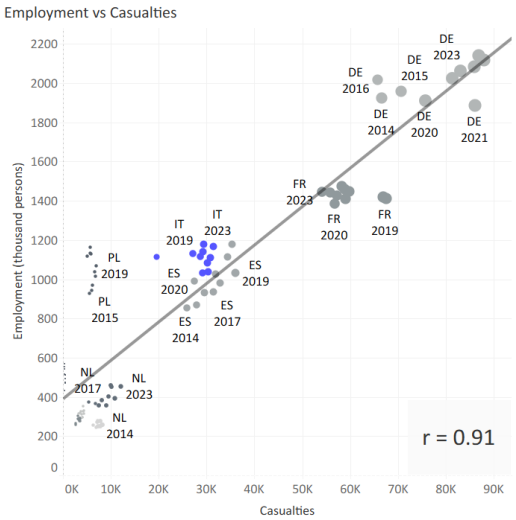
This positioning indicates that the sector combines significant operational exposure with structurally elevated risk conditions associated with logistics activities, vehicle operation, and goods handling.

At the same time, the sector does not represent the most hazardous environment within the Italian economy, as certain industrial activities exhibit higher absolute accident burdens.

Overall, the evidence suggests that Transportation and Storage occupies a mid-to-high position in the national occupational risk distribution, confirming its relevance as a priority area for safety monitoring and prevention policies.

4.4 Exposure and Injury Relationship

Figure 3.20 — Relationship Between Employment Exposure and Recorded Injuries, Top-10 EU Countries, 2014–2023



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

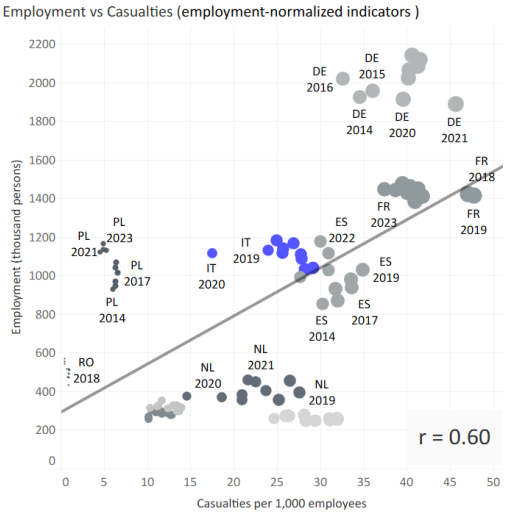
Extending the cross-country comparison presented in Section 4.1, this analysis examines the exposure–injury relationship using annual observations for each country over the period 2014–2023.

The resulting scatter distribution shows a consistent positive association between workforce size and accident counts across time. Countries with larger employment levels in the Transportation and Storage sector systematically record higher numbers of workplace injuries in each year of observation.

The calculated correlation coefficient (r = 0.91) confirms the strength and stability of this relationship across the dataset. The pattern indicates that fluctuations in accident totals largely reflect changes in workforce exposure rather than structural shifts in risk conditions.

This temporal consistency reinforces the interpretation that employment size is the primary driver of variation in absolute accident counts, confirming the analytical necessity of using employment-normalized indicators when comparing occupational risk.

Figure 3.21 — Relationship Between Employment Exposure and Injury Incidence in Transportation and Storage (Top-10 EU Countries, 2014–2023)



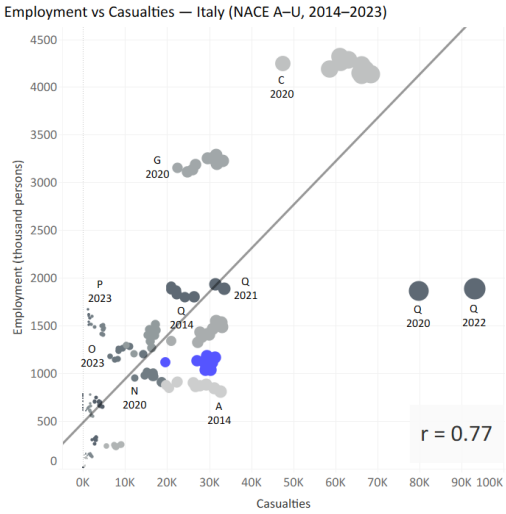
Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

A complementary scatter analysis using employment-normalized indicators (casualties per 1,000 employees) shows a substantially weaker relationship between sector size and injury incidence. When the same country–year observations are compared using normalized rates, the correlation decreases to r = 0.60. This reduction indicates that while absolute accident totals scale strongly with workforce size, relative occupational risk varies more independently across countries. The weaker association confirms that employment-adjusted indicators capture structural differences in safety performance that are not explained solely by exposure levels.

A similar pattern emerges when the exposure–injury relationship is examined across economic activities within Italy (NACE A–U).

The comparison between sectoral employment size and total accident counts shows a relatively strong positive association (r = 0.77), indicating that sectors employing larger numbers of workers tend to report higher numbers of workplace injuries in absolute terms.

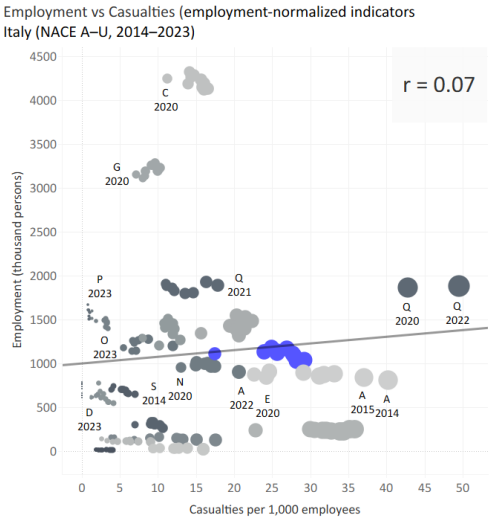
Figure 3.22 — Employment Size and Total Workplace Injuries Across Italian Economic Activities (NACE A–U, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

However, when injury incidence is expressed using employment-normalized indicators (casualties per 1,000 employees), the relationship largely disappears. The correlation between sector size and injury rates falls to r = 0.07, suggesting that the relative occupational risk of a sector is essentially independent of its workforce size.

Figure 3.23 — Employment Size and Injury Incidence Across Italian Economic Activities (Casualties per 1,000 Employees, NACE A–U, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

This contrast confirms that absolute accident totals are primarily driven by exposure levels, whereas normalized indicators capture structural differences in safety conditions across sectors. The finding reinforces the methodological importance of using employment-adjusted metrics when comparing occupational risk across both countries and economic activities.

5. Strategic Insights & Implications

5.1 Key Findings

The exploratory analysis highlights several structural patterns in work-related injuries within the Transportation and Storage sector:

- Temporal pattern: Injury incidence remained broadly stable over the 2014–2023 period, with moderate year-to-year variation but no persistent long-term trend.
- Cross-country heterogeneity: Differences in absolute accident counts across countries largely reflect variations in workforce size, while normalized injury rates show more limited divergence among major sector economies.
- Dominant accident mechanisms: A substantial share of injuries occurs within a limited number of operational contexts associated with transport operations, manual handling activities, and vehicle-related tasks.
- Gender and structural exposure patterns: The sector workforce is strongly male-dominated, which contributes to the higher share of male casualties observed in injury statistics.
- Italy's sector positioning: Within the Italian economy, Transportation and Storage exhibits a higher injury incidence than the national average across economic activities.

5.2 Implications of Observed Patterns

The identified patterns suggest several practical implications for occupational safety analysis and policy design.

- Priority areas for safety investment – Preventive interventions should focus on the operational contexts most frequently associated with accidents, particularly logistics handling activities and vehicle-related operations.
- Structural vs cyclical risks – The relative stability of injury rates suggests that many risks in the sector are structural characteristics of transport and logistics activities rather than short-term cyclical fluctuations.
- Need for deeper analytical modeling – Further investigation may benefit from more granular analysis of operational contexts, occupational roles, and severity outcomes.

5.3 Analytical Limitations

Circumstance classification limitations

Some ESAW circumstance dimensions contain a substantial share of observations classified as "Not specified" or "No information."

In the Italian Transportation and Storage sector, these categories account for a notable proportion of cases in the Workstation and Working Process dimensions. This pattern reflects limitations in the underlying reporting systems rather than meaningful operational mechanisms.

Consequently, interpretations based on these specific dimensions should be treated with caution, and the analysis therefore prioritizes circumstance categories where classification coverage is more complete.

Additional analytical constraints

- The analysis is descriptive rather than causal.
- Age-standardized injury rates could not be constructed due to incompatible age classifications between employment and injury datasets.
- Backcasted values were introduced only to maintain aggregate time-series continuity and are interpreted cautiously.

5.4 Future Analytical Extensions

While age and severity variables are available in the underlying datasets, they were not included in the core analysis because of limited granularity and classification incompatibilities between employment and injury statistics.

Future research could therefore:

- Develop age-specific injury incidence indicators through age-standardized rate calculations.
- Examine severity patterns if more detailed injury classifications become available.
- Integrate occupational or task-level variables to better isolate operational risk drivers.

5.5 Final Takeaway

The analysis indicates that occupational injury risk in the Transportation and Storage sector is largely shaped by structural characteristics of transport and logistics operations rather than by short-term fluctuations.

Cross-country comparisons show broadly comparable risk levels among major European sector economies once employment exposure is considered.

Within Italy, the sector exhibits a higher injury incidence than the national economic average, confirming its classification as a relatively high-risk activity.

Accident patterns are concentrated in a limited number of operational contexts, suggesting clear priorities for preventive interventions.

While the analysis remains descriptive, the curated dataset provides a transparent foundation for more detailed occupational safety research and evidence-based policy discussion.

6. Use of Artificial Intelligence Tools

Artificial intelligence (AI) tools were used as a supporting resource during the preparation of this document. Their use was limited to assisting with drafting, language editing, and the structural organization of the analytical narrative, including suggestions to improve clarity, terminology consistency, and formatting in line with professional analytical reporting standards.

All substantive analytical activities—including data preparation, SQL query development, indicator computation, chart design, and interpretation of results—were independently performed and validated by the author. AI tools were not used to generate, modify, or process the underlying datasets used in the analysis.

The exploratory analysis presented in this report is based on the curated relational dataset described in the accompanying Data Curation document, and all methodological decisions and analytical conclusions reflect the author's independent work.

The author retains full responsibility for the accuracy, interpretation, and conclusions presented in this report.

7. Appendices

Appendix 7.1 – Supplementary Tables

Table A7.1.1 — Average annual employment in NACE H by major country groups and gender composition, 2014–2023

Country	Average Employment (thousand persons)	Employment Share (%)	Female (%)	Male (%)
DE	2,012.94	18.42	24.91	75.09
FR	1,433.00	13.11	25.98	74.02
IT	1,111.92	10.18	20.43	79.57
PL	1,051.28	9.62	20.56	79.44
ES	991.50	9.07	19.9	80.1
RO	508.25	4.65	13.94	86.06
NL	400.19	3.66	23.78	76.22
CZ	318.93	2.92	25.8	74.2
HU	283.10	2.59	24.16	75.84
BE	262.10	2.4	20.71	79.29
Other	2,554.06	23.37	21.53	78.47

Table A7.1.2 — Severity of Workplace Injuries in the Transportation and Storage Sector (NACE H): Fatal and Non-Fatal Accidents, 2014–2023

Year	Fatal Accidents	Non-Fatal Accidents	Total Accidents	Fatal Share
2014	443.0	208,152.0	208,595.0	0.212
2015	463.0	213,270.0	213,733.0	0.217
2016	437.0	211,772.0	212,209.0	0.206
2017	425.0	238,670.0	239,095.0	0.178
2018	439.0	248,340.0	248,779.0	0.176
2019	405.0	250,357.0	250,762.0	0.162
2020	378.0	204,880.0	205,258.0	0.184
2021	444.0	236,167.0	236,611.0	0.188
2022	402.0	233,122.0	233,524.0	0.172
2023	431.0	233,505.0	233,936.0	0.184
2014–2023 Total	4267.0	2,278,235.0	2,282,502.0	0.187

Annual counts of fatal and non-fatal workplace injuries in the Transportation and Storage sector (NACE H), with the corresponding share of fatalities relative to total reported accidents over the 2014–2023 period.

Table A7.1.3 — Average Injury Incidence in Transportation and Storage, NACE H, Across the Top-10 European Economies, 2014–2023

Country	Average Casualties	Average Employment (thousand employees)	Injury Rate (per 1,000 employees)	Rank
France	59413	1433	41.49	1
Germany	78991	2013	39.22	2
Spain	31272	992	31.57	3
Belgium	7414	262	28.34	4
Italy	28596	1112	25.77	5
Netherlands	8932	400	22.21	6
Czechia	3964	319	12.45	7
Hungary	3257	283	11.5	8
Poland	6030	1051	5.78	9
Romania	381	508	0.76	10

Table A7.1.4 — Injury Incidence by Economic Activity in Italy, NACE A–U, Average 2014–2023

Activity	Average Casualties	Average Employment	Injury Rate (per 1,000 employees)	Rank
E - Water supply; sewerage, waste management and remediation activities	7693	241	31.92	1
A - Agriculture, forestry and fishing	25890	871	29.86	2
H - Transportation and storage	28596	1112	25.77	3
F - Construction	29266	1433	20.38	4
Q - Human health and social work activities	37513	1865	20.08	5
N - Administrative and support service activities	16060	976	16.49	6
C - Manufacturing	62525	4212	14.86	7
B - Mining and quarrying	385	31	12.45	8
I - Accommodation and food service activities	15387	1375	11.16	9
L - Real estate activities	1499	146	10.52	10
R - Arts, entertainment and recreation	2875	309	9.33	11
G - Wholesale and retail trade; repair of motor vehicles and motorcycles	28881	3205	9.00	12
O - Public administration and defence; compulsory social security	9277	1224	7.57	13
D - Electricity, gas, steam and air conditioning supply	644	118	5.55	14
S - Other service activities	3645	695	5.28	15
J - Information and communication	1860	619	3.06	16
U - Activities of extraterritorial organisations and bodies	46	17	2.81	17

Activity	Average Casualties	Average Employment	Injury Rate (per 1,000 employees)	Rank
M - Professional, scientific and technical activities	4002	1481	2.72	18
K - Financial and insurance activities	1111	627	1.77	19
P - Education	1466	1584	0.93	20
T - Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	27	710	0.04	21

Table A7.1.5 — Distribution of Accident Circumstances in NACE H, Top-10 Countries, Average 2014–2023

Category	Subcategory	Share within category (%)	Rank
Contact mode of injury	Physical or mental stress	32.12	1
Contact mode of injury	Horizontal or vertical impact with or against a stationary object	28.95	2
Contact mode of injury	Struck by object in motion	18.83	3
Contact mode of injury	Contact with sharp, pointed, rough, coarse Material Agent	7.35	4
Contact mode of injury	Trapped, crushed, etc.	6.53	5
Contact mode of injury	Bite, kick, etc.	3.08	6
Contact mode of injury	Other contacts	1.6	7
Contact mode of injury	Contact with electrical voltage, temperature, hazardous substances	1.47	8
Contact mode of injury	Drowned, buried, enveloped	0.06	9
Deviation	Body movement under or with physical stress	24.57	1
Deviation	Slipping - stumbling and falling - fall of persons	23.87	2
Deviation	Loss of control of machine, means of transport or handling equipment, hand-held tool, object, animal	22.79	3
Deviation	Body movement without any physical stress	15.03	4
Deviation	Shock, fright, violence, aggression, threat, presence	5.86	5
Deviation	Breakage, bursting, splitting, slipping, fall, collapse of material agent	5.32	6
Deviation	Other deviations	1.18	7
Deviation	Deviation by overflow, overturn, leak, flow, vaporisation, emission	1.09	8
Deviation	Deviation due to electrical problems, explosion, fire	0.28	9

Category	Subcategory	Share within category (%)	Rank
Specific physical activity	Movement	32.89	1
Specific physical activity	Driving/being on board a means of transport or handling equipment	19.79	2
Specific physical activity	Carrying by hand	19.42	3
Specific physical activity	Handling of objects	18.12	4
Specific physical activity	Presence	3.46	5
Specific physical activity	Working with hand-held tools	2.89	6
Specific physical activity	Other specific physical activities	2.16	7
Specific physical activity	Operating machine	1.26	8
Working environment	Industrial site	39.42	1
Working environment	Public area	34.49	2
Working environment	No information	13.91	3
Working environment	Tertiary activity area, office, amusement area, miscellaneous	4.07	4
Working environment	In the home	2.22	5
Working environment	Construction site, construction, opencast quarry, opencast mine	1.65	6
Working environment	Other working environments	1.34	7
Working environment	On /over water	1.01	8
Working environment	In the air, elevated	0.86	9
Working environment	Sports area	0.41	10
Working environment	Farming, breeding, fish farming, forest zone	0.38	11
Working environment	Health establishment	0.17	12

Category	Subcategory	Share within category (%)	Rank
Working environment	Underground	0.07	13
Working environment	In high pressure environments	0.01	14
Working process	No information	68.74	1
Working process	Production, manufacturing, processing, storing	11.55	2
Working process	Movement, sport, artistic activity	10.61	3
Working process	Other work processes	4.18	4
Working process	Service, intellectual activity	3.71	5
Working process	Other	0.73	6
Working process	Excavation, construction, repair, demolition	0.34	7
Working process	Agriculture, forestry, horticulture, fish farming, work with live animals	0.14	8
Workstation	Usual workstation or within the usual local unit of work	42.58	1
Workstation	Occasional or mobile workstation	31.5	2
Workstation	Not specified	24.93	3
Workstation	Other workstation	0.99	4

Table A7.1.6 — Gender Composition of Employment: Transportation and Storage vs Total Economy in Italy, Average 2014–2023

Activity	Male Employment (*)	Female Employment (*)	Male Employment Share (%)	Female Employment Share (%)
A-U - Total Economy	13,236.00	9,611.00	57.93	42.07
A - Agriculture, forestry and fishing	641.00	230.00	73.59	26.41
B - Mining and quarrying	27.00	4.00	87.32	12.68
C - Manufacturing	3,099.00	1,113.00	73.58	26.42
D - Electricity, gas, steam and air conditioning supply	88.00	30.00	74.82	25.18
E - Water supply; sewerage, waste management and remediation activities	204.00	37.00	84.71	15.29
F - Construction	1,335.00	99.00	93.11	6.89
G - Wholesale and retail trade; repair of motor vehicles and motorcycles	1,876.00	1,329.00	58.54	41.46
H - Transportation and storage	885.00	227.00	79.57	20.43
I - Accommodation and food service activities	680.00	695.00	49.44	50.56
J - Information and communication	433.00	185.00	70.05	29.95
K - Financial and insurance activities	342.00	285.00	54.51	45.49
L - Real estate activities	80.00	65.00	55.08	44.92
M - Professional, scientific and technical activities	800.00	681.00	53.99	46.01
N - Administrative and support service activities	461.00	514.00	47.29	52.71
O - Public administration and defence; compulsory social security	802.00	421.00	65.57	34.43
P - Education	392.00	1,192.00	24.75	75.25
Q - Human health and social work activities	555.00	1,310.00	29.77	70.23

Activity	Male Employment (*)	Female Employment (*)	Male Employment Share (%)	Female Employment Share (%)
R - Arts, entertainment and recreation	174.00	134.00	56.5	43.5
S - Other service activities	262.00	433.00	37.64	62.36
T - Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	91.00	619.00	12.82	87.18
U - Activities of extraterritorial organisations and bodies	9.00	8.00	52.59	47.41

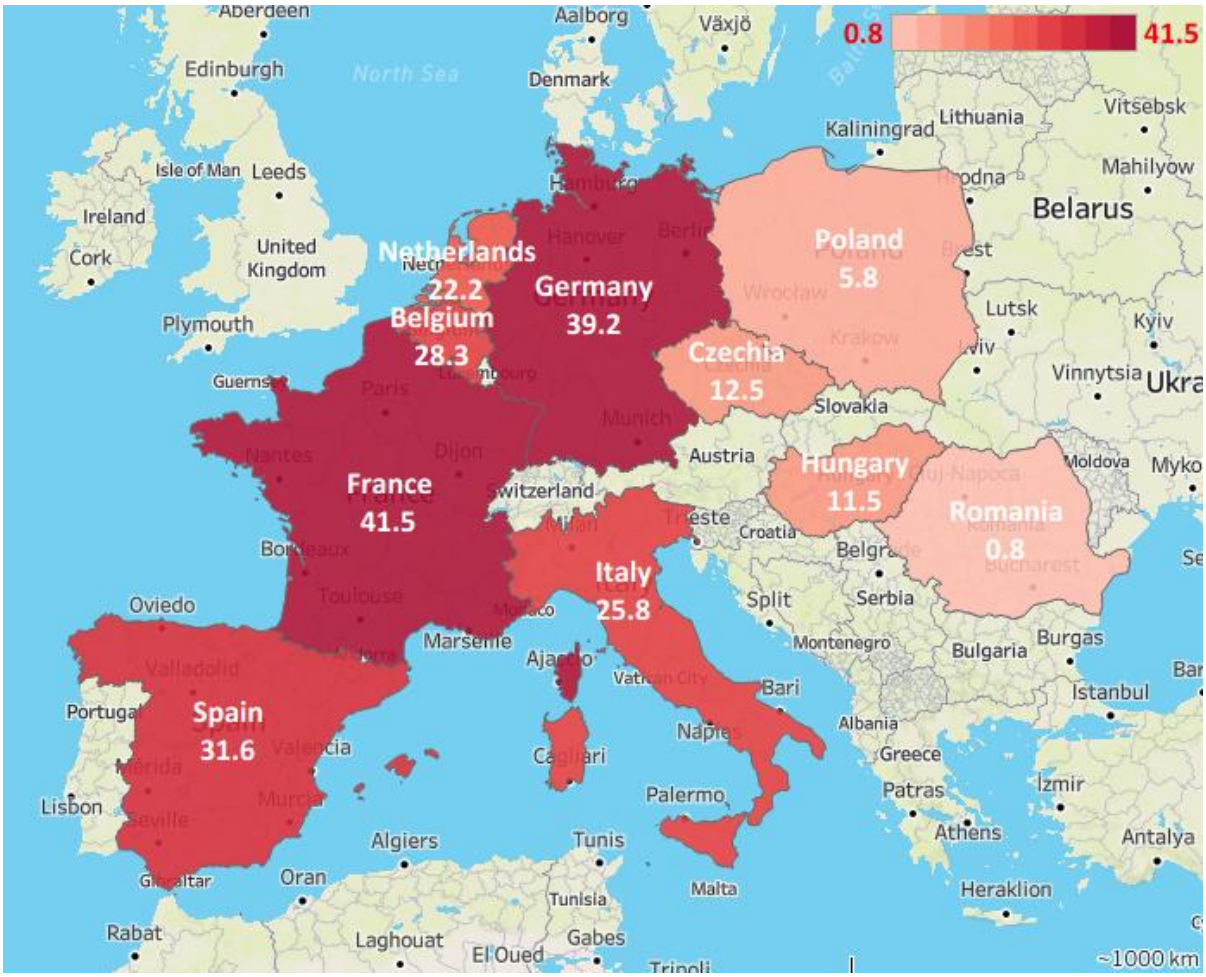
(*) Average Employment (thousand persons)

Appendix 7.2 – Supplementary Charts

Figure A7.2.1 — Average Employment in NACE H, Italy vs Top Sector Economies (2014–2023)

Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

Figure A7.2.2 — Average Injury Incidence (Casualties per 1,000 Employees, 2014–2023)



Source: Eurostat, European Statistics on Accidents at Work (ESAW); author's calculations.

Appendix 7.3 – Reference Classification Tables

This appendix presents the reference classification tables used in the relational database.

These dimension tables standardize the categorical variables derived from EUROSTAT ESAW and Labour Force Survey (LFS) datasets and provide the contextual attributes used to structure the fact tables in the analytical schema.

Table 7.3.1 – Dimension Table: Countries

Purpose: Geographic reference entities.

Country code	Country name
AT	Austria
BE	Belgium
BA	Bosnia and Herzegovina
BG	Bulgaria
CH	Switzerland
CY	Cyprus
CZ	Czechia
DE	Germany
DK	Denmark
EE	Estonia
EL	Greece
ES	Spain
FI	Finland
FR	France
HR	Croatia
HU	Hungary

Country code	Country name
IE	Ireland
IS	Iceland
IT	Italy
LT	Lithuania
LU	Luxembourg
LV	Latvia
ME	Montenegro
MK	North Macedonia
MT	Malta
NL	Netherlands
NO	Norway
PL	Poland
PT	Portugal
RO	Romania
RS	Serbia
SE	Sweden
SI	Slovenia
SK	Slovakia
TR	Turkey
UK	United Kingdom

Table 7.3.2 – Dimension Table: Economic Activities
Purpose: NACE Rev.2 sector classification.

NACE activity code	Activity name
A	Agriculture, forestry and fishing
B	Mining and quarrying
C	Manufacturing
D	Electricity, gas, steam and air conditioning supply
E	Water supply; sewerage, waste management and remediation activities
F	Construction
G	Wholesale and retail trade; repair of motor vehicles and motorcycles
H	Transportation and storage
I	Accommodation and food service activities
J	Information and communication
K	Financial and insurance activities
L	Real estate activities
M	Professional, scientific and technical activities
N	Administrative and support service activities
O	Public administration and defence; compulsory social security
P	Education
Q	Human health and social work activities
R	Arts, entertainment and recreation

NACE activity code	Activity name
S	Other service activities
T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use
U	Activities of extraterritorial organisations and bodies

Table 7.3.3 – Dimension Table: Sex
Purpose: Gender classification.

Gender code	Gender description
F	Females
M	Males
UNK	Unknown

Table 7.3.4 – Dimension Table: Age Classes
Purpose: ESAW and LFS age groups.

Age class code	Age category label
Y_GE15	15 years or over
UNK	Unknown
Y_GE55	55 years or over
Y_LT25	Less than 25 years

Table 7.3.5 – Dimension Table: Injury Severity

Purpose: Classification of accident severity.

Severity code	Severity label
FAT	Fatal Accidents
D_GE4	Non-fatal accidents - 4 days or over

Table 7.3.6 – Dimension Table: Circumstance Categories and Subcategories

Purpose: High-level ESAW accident context classification and detailed ESAW accident modality classification.

Category code	Circumstance group	Subcategory code	Specific circumstance
CMI	Contact mode of injury	CMI80	Bite, kick, etc.
CMI	Contact mode of injury	CMI10	Contact with electrical voltage, temperature, hazardous substances
CMI	Contact mode of injury	CMI50	Contact with sharp, pointed, rough, coarse Material Agent
CMI	Contact mode of injury	CMI20	Drowned, buried, enveloped
CMI	Contact mode of injury	CMI30	Horizontal or vertical impact with or against a stationary object
CMI	Contact mode of injury	CMI90	Other contacts
CMI	Contact mode of injury	CMI70	Physical or mental stress
CMI	Contact mode of injury	CMI40	Struck by object in motion
CMI	Contact mode of injury	CMI60	Trapped, crushed, etc.
DEV	Deviation	DEV_7	Body movement under or with physical stress
DEV	Deviation	DEV_6	Body movement without any physical stress
DEV	Deviation	DEV_3	Breakage, bursting, splitting, slipping, fall, collapse of material agent

Category code	Circumstance group	Subcategory code	Specific circumstance
DEV	Deviation	DEV_2	Deviation by overflow, overturn, leak, flow, vaporisation, emission
DEV	Deviation	DEV_1	Deviation due to electrical problems, explosion, fire
DEV	Deviation	DEV_4	Loss of control of machine, means of transport or handling equipment, hand-held tool, object, animal
DEV	Deviation	DEV_9	Other deviations
DEV	Deviation	DEV_8	Shock, fright, violence, aggression, threat, presence
DEV	Deviation	DEV_5	Slipping - stumbling and falling - fall of persons
PHA	Specific physical activity	PHA_5	Carrying by hand
PHA	Specific physical activity	PHA_3	Driving/being on board a means of transport or handling equipment
PHA	Specific physical activity	PHA_4	Handling of objects
PHA	Specific physical activity	PHA_6	Movement
PHA	Specific physical activity	PHA_1	Operating machine
PHA	Specific physical activity	PHA_9	Other specific physical activities
PHA	Specific physical activity	PHA_7	Presence
PHA	Specific physical activity	PHA_2	Working with hand-held tools
WEN	Working environment	WEN_02	Construction site, construction, opencast quarry, opencast mine
WEN	Working environment	WEN_03	Farming, breeding, fish farming, forest zone
WEN	Working environment	WEN_05	Health establishment
WEN	Working environment	WEN_12	In high pressure environments

Category code	Circumstance group	Subcategory code	Specific circumstance
WEN	Working environment	WEN_09	In the air, elevated
WEN	Working environment	WEN_07	In the home
WEN	Working environment	WEN_01	Industrial site
WEN	Working environment	WEN_00	No information
WEN	Working environment	WEN_11	On /over water
WEN	Working environment	WEN_99	Other working environments
WEN	Working environment	WEN_06	Public area
WEN	Working environment	WEN_08	Sports area
WEN	Working environment	WEN_04	Tertiary activity area, office, amusement area, miscellaneous
WEN	Working environment	WEN_10	Underground
WPR	Working process	WPR3	Agriculture, forestry, horticulture, fish farming, work with live animals
WPR	Working process	WPR2	Excavation, construction, repair, demolition
WPR	Working process	WPR6	Movement, sport, artistic activity
WPR	Working process	NINFO	No information
WPR	Working process	OTH	Other
WPR	Working process	WPR5	Other work processes
WPR	Working process	WPR1	Production, manufacturing, processing, storing
WPR	Working process	WPR4	Service, intellectual activity
WST	Workstation	WST_0	Not specified

Category code	Circumstance group	Subcategory code	Specific circumstance
WST	Workstation	WST_2	Occasional or mobile workstation
WST	Workstation	WST_9	Other workstation
WST	Workstation	WST_1	Usual workstation or within the usual local unit of work

Appendix 7.4 — Key Performance Indicators (KPIs)

This appendix summarizes the principal quantitative indicators derived from the exploratory analysis.

Unless otherwise specified, indicators refer to average values over the period 2014–2023.

Table 7.4.1 – Employment Structure — Cross-Country Context (Section 3.1)

Indicator	Definition	Value
Employment Exposure Concentration	Top-10 employment share of total	76.6 %
Sector Employment Growth Index	Employment index (2014=100)	116.4
Gender Employment Ratio	Male/Female employment ratio	3.8
Female Workforce Share	Female employment share	20.9 %

Table 7.4.2 – Injury Incidence — Cross-Country Comparison (Section 3.2)

Indicator	Definition	Value
Peer-Group Average Injury Incidence	Average annual workplace injuries per 1,000 employees across the Top-10 EU countries	21.9
Italy Injury Incidence	Average annual workplace injuries per 1,000 employees in Italy	25.8
Italy Relative Risk vs Peer Average	Ratio between Italy's incidence rate and the peer-country average	1.18

Table 7.4.3 – Injury Incidence — Cross-Sector Comparison in Italy (Section 3.3)

Indicator	Definition	Value
Italy Injury Incidence	Avg casualties per 1,000 employees in Transportation and Storage	25.8
Total Economy Injury Incidence	Avg casualties per 1,000 employees across Italy A–U	12.2
Relative Risk Index	NACE H rate ÷ Total economy rate	2.1

Note: Total Economy Injury Incidence is computed as an employment-weighted aggregate of sectoral injury rates across activities A–U, ensuring that each sector contributes proportionally to its workforce size.

Table 7.4.4 – Accident Circumstance Structure (Section 3.4)

Indicator	Definition
No KPI is reported for this section	Circumstance datasets represent distributions of accident characteristics rather than additive totals. Structural patterns are therefore interpreted directly from the percentage distributions presented in Section 3.4.

Table 7.4.5 – Employment Context of NACE H in Italy (Section 3.5)

Indicator	Definition	Value
Sector Employment Share (NACE H)	Share of Italian employment represented by NACE H	4.9 %
Gender Imbalance Index (Total Economy)	Ratio of male to female employment across Italy A–U	1.4
Male Dominance Ratio (NACE H)	Ratio of male to female employment in NACE H	3.9